Task 2 : Perform SQL queries on the table

Create Database:

```
create database DataSql;
use datasql;
```

Create Table :

```
CREATE TABLE games (
    game_id INT PRIMARY KEY,
    game_name VARCHAR(100),
    genre VARCHAR(50),
    platform VARCHAR(50),
    developer VARCHAR(100),
    publisher VARCHAR(100),
    release_year INT,
    price DECIMAL(10,2),
    rating DECIMAL(3,1),
    sales_millions DECIMAL(10,2)
);
```

Insert into table:

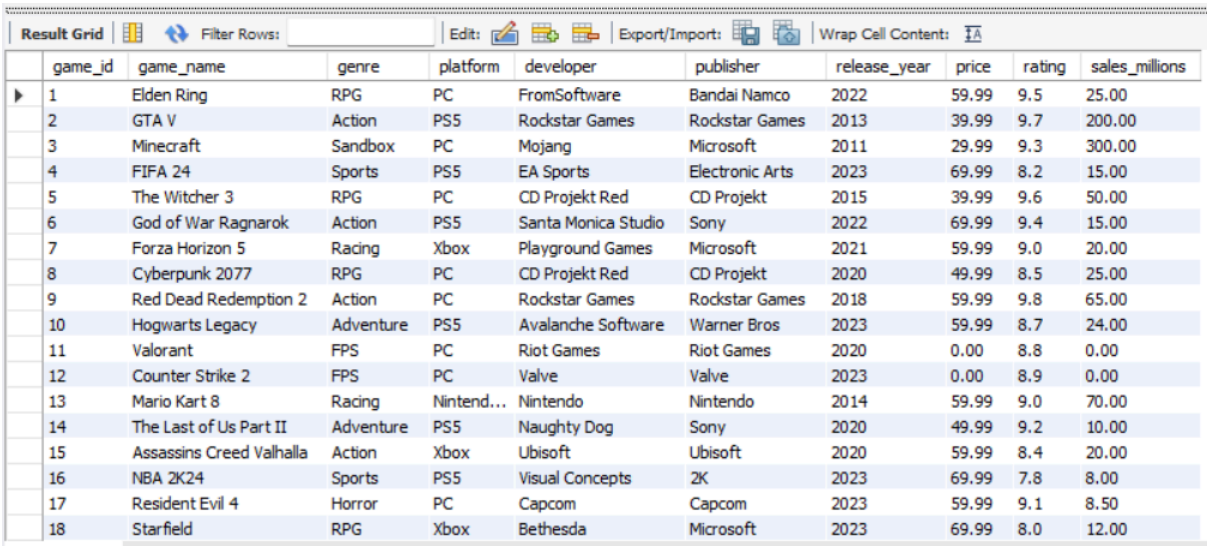
```
INSERT INTO games
(game_id, game_name, genre, platform, developer, publisher, release_year, price, rating,
sales_millions)
VALUES
(1, 'Elden Ring', 'RPG', 'PC', 'FromSoftware', 'Bandai Namco', 2022, 59.99, 9.5, 25.00),
(2, 'GTA V', 'Action', 'PS5', 'Rockstar Games', 'Rockstar Games', 2013, 39.99, 9.7, 200.00),
(3, 'Minecraft', 'Sandbox', 'PC', 'Mojang', 'Microsoft', 2011, 29.99, 9.3, 300.00),
(4, 'FIFA 24', 'Sports', 'PS5', 'EA Sports', 'Electronic Arts', 2023, 69.99, 8.2, 15.00),
(5, 'The Witcher 3', 'RPG', 'PC', 'CD Projekt Red', 'CD Projekt', 2015, 39.99, 9.6, 50.00),
(6, 'God of War Ragnarok', 'Action', 'PS5', 'Santa Monica Studio', 'Sony', 2022, 69.99, 9.4,
15.00),
(7, 'Forza Horizon 5', 'Racing', 'Xbox', 'Playground Games', 'Microsoft', 2021, 59.99, 9.0,
20.00),
(8, 'Cyberpunk 2077', 'RPG', 'PC', 'CD Projekt Red', 'CD Projekt', 2020, 49.99, 8.5, 25.00),
(9, 'Red Dead Redemption 2', 'Action', 'PC', 'Rockstar Games', 'Rockstar Games', 2018,
59.99, 9.8, 65.00),
(10, 'Hogwarts Legacy', 'Adventure', 'PS5', 'Avalanche Software', 'Warner Bros', 2023, 59.99,
8.7, 24.00),
(11, 'Valorant', 'FPS', 'PC', 'Riot Games', 'Riot Games', 2020, 0.00, 8.8, 0.00),
(12, 'Counter Strike 2', 'FPS', 'PC', 'Valve', 'Valve', 2023, 0.00, 8.9, 0.00),
(13, 'Mario Kart 8', 'Racing', 'Nintendo Switch', 'Nintendo', 'Nintendo', 2014, 59.99, 9.0,
70.00),
(14, 'The Last of Us Part II', 'Adventure', 'PS5', 'Naughty Dog', 'Sony', 2020, 49.99, 9.2,
10.00),
(15, 'Assassins Creed Valhalla', 'Action', 'Xbox', 'Ubisoft', 'Ubisoft', 2020, 59.99, 8.4, 20.00),
```

(19, 'Halo Infinite', 'FPS', 'Xbox', '343 Industries', 'Microsoft', 2021, 59.99, 8.3, 10.00),
(20, 'Animal Crossing New Horizons', 'Simulation', 'Nintendo Switch', 'Nintendo', 'Nintendo', 2020, 59.99, 9.1, 45.00);

Queries:

1. **Select Query:**

   select * from games;



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

game_id	game_name	genre	platform	developer	publisher	release_year	price	rating	sales_millions
1	Elden Ring	RPG	PC	FromSoftware	Bandai Namco	2022	59.99	9.5	25.00
2	GTA V	Action	PS5	Rockstar Games	Rockstar Games	2013	39.99	9.7	200.00
3	Minecraft	Sandbox	PC	Mojang	Microsoft	2011	29.99	9.3	300.00
4	FIFA 24	Sports	PS5	EA Sports	Electronic Arts	2023	69.99	8.2	15.00
5	The Witcher 3	RPG	PC	CD Projekt Red	CD Projekt	2015	39.99	9.6	50.00
6	God of War Ragnarok	Action	PS5	Santa Monica Studio	Sony	2022	69.99	9.4	15.00
7	Forza Horizon 5	Racing	Xbox	Playground Games	Microsoft	2021	59.99	9.0	20.00
8	Cyberpunk 2077	RPG	PC	CD Projekt Red	CD Projekt	2020	49.99	8.5	25.00
9	Red Dead Redemption 2	Action	PC	Rockstar Games	Rockstar Games	2018	59.99	9.8	65.00
10	Hogwarts Legacy	Adventure	PS5	Avalanche Software	Warner Bros	2023	59.99	8.7	24.00
11	Valorant	FPS	PC	Riot Games	Riot Games	2020	0.00	8.8	0.00
12	Counter Strike 2	FPS	PC	Valve	Valve	2023	0.00	8.9	0.00
13	Mario Kart 8	Racing	Nintend...	Nintendo	Nintendo	2014	59.99	9.0	70.00
14	The Last of Us Part II	Adventure	PS5	Naughty Dog	Sony	2020	49.99	9.2	10.00
15	Assassins Creed Valhalla	Action	Xbox	Ubisoft	Ubisoft	2020	59.99	8.4	20.00
16	NBA 2K24	Sports	PS5	Visual Concepts	2K	2023	69.99	7.8	8.00
17	Resident Evil 4	Horror	PC	Capcom	Capcom	2023	59.99	9.1	8.50
18	Starfield	RPG	Xbox	Bethesda	Microsoft	2023	69.99	8.0	12.00

2. **Select specific columns**

   SELECT game_name, genre, price
   FROM games;

game_name	genre	price
Elden Ring	RPG	59.99
GTA V	Action	39.99
Minecraft	Sandbox	29.99
FIFA 24	Sports	69.99
The Witcher 3	RPG	39.99
God of War Ragnarok	Action	69.99
Forza Horizon 5	Racing	59.99
Cyberpunk 2077	RPG	49.99
Red Dead Redemption 2	Action	59.99
Hogwarts Legacy	Adventure	59.99
Valorant	FPS	0.00
Counter Strike 2	FPS	0.00
Mario Kart 8	Racing	59.99
The Last of Us Part II	Adventure	49.99
Assassins Creed Valhalla	Action	59.99
NBA 2K24	Sports	69.99
Resident Evil 4	Horror	59.99
Starfield	RPG	69.99

3. **Find games with price greater than $60**

```
SELECT *
FROM games
WHERE price > 50;
```

game_id	game_name	genre	platform	developer	publisher	release_year	price	rating	sales_millions
4	FIFA 24	Sports	PS5	EA Sports	Electronic Arts	2023	69.99	8.2	15.00
6	God of War Ragnarok	Action	PS5	Santa Monica Studio	Sony	2022	69.99	9.4	15.00
16	NBA 2K24	Sports	PS5	Visual Concepts	2K	2023	69.99	7.8	8.00
18	Starfield	RPG	Xbox	Bethesda	Microsoft	2023	69.99	8.0	12.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. **Find RPG games**

```
SELECT *
FROM games
WHERE genre = 'RPG';
```

game_id	game_name	genre	platform	developer	publisher	release_year	price	rating	sales_millions
1	Elden Ring	RPG	PC	FromSoftware	Bandai Namco	2022	59.99	9.5	25.00
5	The Witcher 3	RPG	PC	CD Projekt Red	CD Projekt	2015	39.99	9.6	50.00
8	Cyberpunk 2077	RPG	PC	CD Projekt Red	CD Projekt	2020	49.99	8.5	25.00
18	Starfield	RPG	Xbox	Bethesda	Microsoft	2023	69.99	8.0	12.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. **Sort games by price**

```
SELECT game_name, price
FROM games
ORDER BY price DESC;
```

game_name	price
God of War Ragnarok	69.99
Starfield	69.99
FIFA 24	69.99
NBA 2K24	69.99
Mario Kart 8	59.99
Assassins Creed Valhalla	59.99
Elden Ring	59.99
Resident Evil 4	59.99
Halo Infinite	59.99
Animal Crossing New Horizons	59.99
Hogwarts Legacy	59.99
Forza Horizon 5	59.99
Red Dead Redemption 2	59.99
The Last of Us Part II	49.99
Cyberpunk 2077	49.99
The Witcher 3	39.99
GTA V	39.99
Minecraft	29.99

6. **Display the top 5 highest-selling games**

```
SELECT game_name, sales_millions
FROM games
ORDER BY sales_millions DESC
```

```
LIMIT 5;
```

game_name	sales_millions
Minecraft	300.00
GTA V	200.00
Mario Kart 8	70.00
Red Dead Redemption 2	65.00
The Witcher 3	50.00

7. **Find average game price**

```
SELECT AVG(price) AS average_price
FROM games;
```

average_price
51.491000

8. **Find highest and lowest rating**

```
SELECT
   MAX(rating) AS highest_rating,
   MIN(rating) AS lowest_rating
FROM games;
```

highest_rating	lowest_rating
9.8	7.8

9. **Number of games in each genre**

```
SELECT
   genre,
   COUNT(*) AS number_of_games
FROM games
GROUP BY genre;
```

genre	number_of_games
Action	4
Adventure	2
FPS	3
Horror	1
Racing	2
RPG	4
Sandbox	1
Simulation	1
Sports	2

10. **Total sales by platform**

```
SELECT
   platform,
   SUM(sales_millions) AS total_sales
FROM games
GROUP BY platform
ORDER BY total_sales DESC;
```

platform	total_sales
PC	473.50
PS5	272.00
Nintendo Switch	115.00
Xbox	62.00

11. **Genres having more than 2 game**

```
SELECT
   genre,
   COUNT(*) AS number_of_games
FROM games
GROUP BY genre
HAVING COUNT(*) > 2;
```

genre	number_of_games
Action	4
FPS	3
RPG	4

12. **Platforms with total sales above 50 million**

```
SELECT
   platform,
   SUM(sales_millions) AS total_sales
FROM games
GROUP BY platform
HAVING SUM(sales_millions) > 50
ORDER BY total_sales DESC;
```

platform	total_sales
PC	473.50
PS5	272.00
Nintendo Switch	115.00
Xbox	62.00

13. For Join : create another one table

```
CREATE TABLE developers (
    developer_id INT PRIMARY KEY,
    developer_name VARCHAR(100),
    country VARCHAR(50),
    founded_year INT
);
```

Insert values into the table:

```
INSERT INTO developers
        (developer_id, developer_name, country, founded_year)
        VALUES
        (1, 'FromSoftware', 'Japan', 1986),
        (2, 'Rockstar Games', 'USA', 1998),
        (3, 'Mojang', 'Sweden', 2009),
        (4, 'EA Sports', 'USA', 1991),
        (5, 'CD Projekt Red', 'Poland', 2002),
        (6, 'Santa Monica Studio', 'USA', 1999),
        (7, 'Playground Games', 'UK', 2010),
        (8, 'Riot Games', 'USA', 2006),
        (9, 'Valve', 'USA', 1996),
        (10, 'Nintendo', 'Japan', 1985),
        (11, 'Naughty Dog', 'USA', 1984),
        (12, 'Ubisoft', 'France', 1986),
        (13, 'Visual Concepts', 'USA', 1988),
        (14, 'Capcom', 'Japan', 1979),
        (15, 'Bethesda', 'USA', 1986),
        (16, '343 Industries', 'USA', 2007),
        (17, 'Avalanche Software', 'USA', 1995);
```

Select to view values:

```
        SELECT * FROM developers;
```

developer_id	developer_name	country	founded_year
1	FromSoftware	Japan	1986
2	Rockstar Games	USA	1998
3	Mojang	Sweden	2009
4	EA Sports	USA	1991
5	CD Projekt Red	Poland	2002
6	Santa Monica Studio	USA	1999
7	Playground Games	UK	2010
8	Riot Games	USA	2006
9	Valve	USA	1996
10	Nintendo	Japan	1985
11	Naughty Dog	USA	1984

14. INNER JOIN

```
SELECT
    g.game_name,
    g.genre,
    g.price,
    d.developer_name,
    d.country
FROM games g
INNER JOIN developers d
    ON g.developer = d.developer_name;
```

game_name	genre	price	developer_name	country
Elden Ring	RPG	59.99	FromSoftware	Japan
GTA V	Action	39.99	Rockstar Games	USA
Minecraft	Sandbox	29.99	Mojang	Sweden
FIFA 24	Sports	69.99	EA Sports	USA
The Witcher 3	RPG	39.99	CD Projekt Red	Poland
God of War Ragnarok	Action	69.99	Santa Monica Studio	USA
Forza Horizon 5	Racing	59.99	Playground Games	UK
Cyberpunk 2077	RPG	49.99	CD Projekt Red	Poland
Red Dead Redemption 2	Action	59.99	Rockstar Games	USA
Hogwarts Legacy	Adventure	59.99	Avalanche Software	USA
Valorant	FPS	0.00	Riot Games	USA

15. Find games developed in Japan

```
SELECT
    g.game_name,
    g.genre,
    d.developer_name,
```

```
    d.country
FROM games g
INNER JOIN developers d
    ON g.developer = d.developer_name
WHERE d.country = 'Japan';
```

game_name	genre	developer_name	country
Elden Ring	RPG	FromSoftware	Japan
Mario Kart 8	Racing	Nintendo	Japan
Resident Evil 4	Horror	Capcom	Japan
Animal Crossing New Horizons	Simulation	Nintendo	Japan

16. Show developer-wise game count

```
SELECT
    d.developer_name,
    d.country,
    COUNT(g.game_id) AS number_of_games
FROM developers d
INNER JOIN games g
    ON d.developer_name = g.developer
GROUP BY d.developer_name, d.country
ORDER BY number_of_games DESC;
```

developer_name	country	number_of_games
CD Projekt Red	Poland	2
Nintendo	Japan	2
Rockstar Games	USA	2
Naughty Dog	USA	1
343 Industries	USA	1
Valve	USA	1
Capcom	Japan	1
Mojang	Sweden	1
Avalanche Software	USA	1
Ubisoft	France	1
FromSoftware	Japan	1

17. **Total sales by developer**

```
SELECT
    d.developer_name,
    d.country,
    SUM(g.sales_millions) AS total_sales
FROM developers d
INNER JOIN games g
    ON d.developer_name = g.developer
GROUP BY d.developer_name, d.country
ORDER BY total_sales DESC;
```

developer_name	country	total_sales
Mojang	Sweden	300.00
Rockstar Games	USA	265.00
Nintendo	Japan	115.00
CD Projekt Red	Poland	75.00
FromSoftware	Japan	25.00
Avalanche Software	USA	24.00
Playground Games	UK	20.00
Ubisoft	France	20.00
Santa Monica Studio	USA	15.00
EA Sports	USA	15.00
Bethesda	USA	12.00